

# Project 3: Discrete logistic growth

## MCSB Bootcamp — Mathematical and Computational Track

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### Model statement

Suppose a rabbit colony has a population  $x(n)$  at month  $n$ , where  $x$  is measured in thousands. If the population were growing in an unbounded environment, the population obeys

$$x(n+1) = x(n) + r \cdot x(n)$$

where  $r$  is the per-capita growth rate. Suppose instead the population is in a bounded environment (like an island), so growth is limited and the population obeys

$$x(n+1) = x(n) + r \left(1 - \frac{x(n)}{K}\right) x(n)$$

where  $K$  is a parameter we refer to as the carrying capacity.

### Part I

Suppose  $r = 0.1$  and  $K = 0.6$ .

- a. According to your intuition, what population sizes are *steady states*, meaning that if the population had that value at time  $n = 0$ , then it would remain at that value?

```
# Your code here.
```

- b. Sketch your intuition for the population  $x(t)$  from a starting population  $x(1) = 0.2$ .

```
# Your code here.
```

## Part II

Write code to solve the dynamical system, and answer the following questions:

- c. Suppose  $r = 0.1$  and  $K = 0.6$ . Generate time series of the populations for a few starting populations  $x(1)$ . Does it match your intuition?

```
# Your code here.
```

- d. Suppose  $r = 2.1$  and  $K = 0.6$ . Generate time series of the populations for a few starting populations  $x(1)$ .

```
# Your code here.
```

## Part III

In a discrete-time dynamical system, if the population cycles between two values, the solution is called a two-cycle. Cycling between  $N$  values is called an  $N$ -cycle.

- e. Check that at  $r = 2.5$  and  $K = 0.6$  there is a 4-cycle.

```
# Your code here.
```

- f. (Optional) Can you find a value of  $r$ ,  $K$  and  $x(1)$  that gives a 3-cycle?

```
# Your code here.
```

- g. In this part, we will do a parameter sweep for  $0 < r < 3.0$ , with fixed  $K = 0.6$ . The goal is to generate a diagram where the horizontal axis is the parameter value  $r$ . On the vertical axis, if there is a stable steady state, plot the steady-state population. If there is an  $N$ -cycle, plot the  $N$  values of  $x$  that it cycles through.

Hint: One way to plot the steady state or the  $N$ -cycle is to simulate the system until  $n_{\max}$ , and plot the last half of the values of  $x(n)$ . You need to choose  $n_{\max}$  large enough so that the dynamics have settled into their steady state (or steady cycle) by  $n_{\max}/2$ .

Hint: How many  $r$  values should you explore?

```
# Your code here.
```

This type of behavior in a dynamical system is called *chaos*! This particular type of chaos is called period-doubling chaos.